Particle Physics Beyond the Standard Model

PHYS:5905

The University of Iowa
The College of Liberal Arts and Sciences
Fall, 2025

Course meeting time and place
11:00 am -- 12:15 pm Tuesday VAN 358
11:00 am -- 12:15 pm Thursdays VAN 358

Department of Physics & Astronomy
https://physics.uiowa.edu/

Instructor
Prof. Matheus Hostert

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319-335-2943

Email
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Office location
VAN 502

Student drop-in hours
2:00 pm -- 3:30 pm Tuesday
2:00 pm -- 3:30 pm Thursday
Or by appointment

Departmental Executive Officer (DEO):
Gregory G. Howes (gregory-howes@uiowa.edu)

Student Complaints
Students with a complaint about a grade or a related matter should first discuss the situation with the instructor and/or the course supervisor (if applicable), and finally with the DEO (Chair) of the department, school or program offering the course. Sometimes students will be referred to the department or program's Director of Undergraduate Studies (DUS) or Director of Graduate Studies (DGS).

Undergraduate students should contact CLAS Undergraduate Programs for support when the matter is not resolved at the previous level. Graduate students should contact the CLAS Graduate Affairs Manager when additional support is needed.

Course's College (Administrative Home)

For undergraduate courses

The College of Liberal Arts and Sciences (CLAS) is the home of this course, and CLAS governs the add and drop deadlines, academic misconduct policies, and other undergraduate policies and procedures. Other UI colleges may have different policies.

For graduate courses

The College of Liberal Arts and Sciences (CLAS) is the home of this course, and CLAS governs the policies and procedures for its courses. Graduate students, however, must adhere to the academic deadlines set by the Graduate College.

Drop Deadline for this Course

You may drop an individual course before the drop deadline; after this deadline you will need collegiate approval. You can look up the drop deadline for this course here. When you drop a course, a "W" will appear on your transcript. The mark of "W" is a neutral mark that does not affect your GPA. To discuss how dropping (or staying in) a course might affect your academic goals, please contact your Academic Advisor. Directions for adding or dropping a course and other registration changes can be found on the Registrar's website. Undergraduate students can find policies on dropping CLAS courses here. Graduate students should adhere to the academic deadlines and policies set by the Graduate College.

UI Email

Students are responsible for all official correspondences sent to their UI email address (uiowa.edu) and must use this address for any communication with instructors or staff in the UI community. For the privacy and the protection of student records, UI faculty and staff can only correspond with UI email addresses.

Description of Course

This graduate course on particle physics beyond the Standard Model (BSM) is designed for junior graduate and senior undergraduate students. Prerequisites are special relativity and undergraduate quantum mechanics. No prior quantum field theory is required.

We begin with a brief review of Standard Model particle content and interactions, then cover decay rates, cross sections, Fermi's Golden Rule, Breit–Wigner propagators, and a practical introduction to Feynman rules sufficient for naive dimensional analysis. We then study neutrino oscillations, wave packets, matter effects, and mass mechanisms. Next, we cover dark matter theories, focusing on thermal freeze-out, freeze-in, and their experimental status. Time permitting, we conclude with a broader survey of the BSM landscape.

The course features in-person blackboard lectures, guided literature reviews, special seminars, and student-led presentations.

LaTeX is encouraged for assignments (templates will be provided on ICON).

We will stick to the one and only metric signature: (+, -, -, -).

Learning Outcomes

By the end of the course, students will be able to:

- Work with natural units.
- Understand the concept of conservation laws, allowed and forbidden processes, and track quantum numbers.
- Interpret Feynman diagrams and apply naive dimensional analysis to estimate basic decay rates and scattering cross sections.
- Calculate neutrino oscillation probabilities in vacuum using plane waves and wave packets.
- Calculate neutrino oscillations in matter, understand the MSW resonance and adiabatic solar neutrino flavor transitions.
- Solve simplified Boltzmann equations for dark matter thermal freeze-out and freeze-in; estimate relic abundances.
- Evaluate strategies and limitations of direct, indirect, and collider dark-matter searches; read experimental exclusion plots.
- Read simple scientific papers on neutrinos and dark matter and write literature reviews.
- Communicate ideas in particle physics clearly in writing and oral presentation.

Textbooks/Materials

No required texts. All required reading will be provided via the University of Iowa Libraries' electronic subscriptions or posted on the course website/ICON.

Course ICON site

To access the course site, log into [Iowa Courses Online (ICON)] using your Hawk ID and password.

Grading System

Grading for this course is Satisfactory/Unsatisfactory (S/U).

Course Grades
Class Attendance and Participation: 10%
Homework: 40%
Final Project (paper + presentation): 50%

Late work: Assignments up to 72 hours late are accepted with a 10% deduction per 24 hours; beyond 72 hours, late work is accepted only with documented circumstances. If you anticipate a conflict, contact the instructor as early as possible to arrange alternatives consistent with University policy.

Attendance will be taken during class.

Homework will be assigned bi-weekly to a total of 5 assignments. Homework comprises brief literature reviews and analytical calculations with optional coding tasks. The lowest homework score will be dropped.

Final Project will comprise of a short group presentation based on a selection of pre-specified topics. Groups will also submit a 6 to 8 page literature review and individual statements of contribution.

CLAS guidance: no large assignments due the week before finals week; this course complies.

Attendance and Accommodations
Regular attendance is required and expected. A total of 10% of your final grade is determined by attendance records. In serious and unforeseen circumstances, you may be excused from attending a class provided you inform the instructor as soon as possible.

<u>Absences from Class</u>
University regulations require that students be allowed to make up examinations which have been missed due to illness, religious holy days, military service obligations, including service-related medical appointments, jury duty, or other unavoidable circumstances or other university-sponsored activities. Students should work with their instructor regarding making up other missed work, such as assignments, quizzes, and classroom attendance.

<u>Absences for Religious Holy Days</u>
The university is prepared to make reasonable accommodations for students whose religious holy days coincide with their classroom assignments, test schedules, and classroom attendance expectations. Students must notify their instructors in writing of any

such religious holy day conflicts or absences within the first few days of the semester or session, and no later than the third week of the semester. If the conflict or absence will occur within the first three weeks of the semester, the student should notify the instructor as soon as possible. See Policy Manual 8.2 Absences for Religious Holy Days for additional information.

Absences for Military Service Obligations

Students absent from class or class-related requirements due to U.S. veteran or U.S. military service obligations (including military service–related medical appointments, military orders, and National Guard Service obligations) shall be excused without any grading adjustment or other penalty. Instructors shall make reasonable accommodations to allow students to make up, without penalty, tests and assignments they missed because of veteran or military service obligations. Reasonable accommodations may include making up missed work following the service obligation; completing work in advance; completing an equivalent assignment; or waiver of the assignment without penalty. In all instances, students bear the responsibility to communicate with their instructors about such veteran or military service obligations, to meet course expectations and requirements.

Accommodations for Students with Disabilities

The University is committed to providing an educational experience that is accessible to all students. If a student has a diagnosed disability or other disabling condition that may impact the student's ability to complete the course requirements as stated in the syllabus, the student may seek accommodations through Student Disability Services (SDS). SDS is responsible for making Letters of Accommodation (LOA) available to the student. **The student must provide an LOA to the instructor as early in the semester as possible, but requests not made at least two weeks prior to the scheduled activity for which an accommodation is sought may not be accommodated.** The LOA will specify what reasonable course accommodations the student is eligible for and those the instructor should provide. Additional information can be found on the SDS website.

Students should feel free to get in touch via email or during office hours to discuss any accommodations necessary, as outlined above.

Other Expectations of Student Performance

Free Speech and Expression

The University of Iowa supports and upholds the First Amendment protection of freedom of speech and the principles of academic and artistic freedom. We are committed to open inquiry, vigorous debate, and creative expression inside and outside of the classroom. Visit the Free Speech at Iowa website for more information on the university's policies on free speech and academic freedom.

<u>Non-discrimination Statement</u>

The University of Iowa prohibits discrimination in employment, educational programs, and activities on the basis of race, creed, color, religion, national origin, age, sex, pregnancy (including childbirth and related conditions), disability, genetic information, status as a U.S. veteran, service in the U.S. military, sexual orientation, or associational preferences. The university also affirms its commitment to providing equal opportunities and equal access to university facilities. For additional information on nondiscrimination policies, contact the Senior Director, Office of Civil Rights Compliance, the University of Iowa, 202 Jessup Hall, Iowa City, IA 52242-1316, 319-335-0705, ui-ocrc@uiowa.edu. Although not required, students have the option to share their pronouns and chosen/preferred names in class and through MyUI. Instructors and advisors can find information about a student's chosen/preferred name in MyUI.

<u>Classroom Expectations</u>

Students are expected to comply with University policies regarding appropriate classroom behavior as outlined in the Code of Student Life. While students have the right to express themselves and participate freely in class, it is expected that students will behave with the same level of courtesy and respect in the virtual class setting (whether asynchronous or synchronous) as they would in an in-person classroom. Failure to follow behavior expectations as outlined in the Code of Student Life may be addressed by the instructor and may also result in discipline under the Code of Student Life policies governing E.5 Disruptive Behavior or E.6 Failure to Comply with University Directive.

<u>Class Recordings</u>

The unauthorized video or audio recording of academic activities (e.g., lectures, course discussions, office hours, etc.) by a student is prohibited. Students with a reasonable accommodation for recording approved by Student Disability Services should notify each instructor and provide the Letter of Accommodation prior to using the accommodation. A student may record classroom activities with prior written permission from the instructor and notice to other students in the class that audio or video recording may occur. Any and all classroom recording must be for personal academic use only. The distribution, sharing, sale, or posting of recordings on the internet (including social media), in whole or in part, is prohibited and doing so may be a violation of the Code of Student Life and/or state or federal privacy, copyright, or other laws.

Academic Honesty and Misconduct

All students in CLAS courses are expected to abide by the college's standards of academic honesty. Undergraduate academic misconduct must be reported by instructors to CLAS according to these procedures. Graduate academic misconduct must be reported to the Graduate College according to Section F of the Graduate College Manual.

Collaborative work is encouraged, but plagiarism is not. Students are responsible for ensuring their own academic honesty and encouraged to ask their instructor for clarification if needed.

Artificial Intelligence (AI) Policy

Usage of AI tools such as ChatGPT, Copilot, or Claude, is allowed and encouraged, especially for coding and idea generation, but it should not be used blindly as hallucinations are common and can look convincing. **AI tools should not be used for generating solutions or code you do not understand**. If AI is used, include a brief "AI usage" note.

Collaboration on ideas is encouraged. However, all submitted work (including code) must be authored and understood by you. Cite any shared sources or discussions.

Date and Time of the Final Exam

This course does not have a final exam.

Calendar of Course Assignments and Exams

Topics Covered	**Dates**
Module 1: Introduction and Review	
Natural units, special relativity, simplified treatment of Feynman Diagrams, Fermi's golden rule, decay rates, cross sections, and the Breit-Wigner propagator.	Weeks 1 – 3 (Aug 26 – Sep 11)
Module 2: Neutrino Physics	
Neutrino sources and interactions, neutrino oscillations, wave packets, matter effects, and direct mass measurements.	Weeks 4 – 7 (Sep 16 – Oct 9)
Module 3: Dark Matter	
Evidence for dark matter, thermal dark matter paradigm, cosmological production of dark matter particles, freeze-out and freeze-in, and experimental signatures.	Weeks 8 – 11 (Oct 14 – Nov 6)
Module 4: Final Project	

Short group presentation based on a selection of pre-specified topics and a 6 to 8 page literature review.	Weeks 12 and 13 (Nov 11 – Nov 20)
Fall Break (no classes November 24-28)	
Module 5: Special Interests	
Survey and discussions of other scenarios of beyond-the-Standard-Model theories and their current status.	Week 14 and 15 (Dec 02 – Dec 11)

Student Support Resources and Related Policies

Academic Support for this Course

I am available for help during drop-in hours, via email, or by appointment. Please feel free to reach out. You may also access academic support via Tutor Iowa at https://tutor.uiowa.edu/ and writing support via The Writing Center at https://writingcenter.uiowa.edu/.

Mental Health

Students are encouraged to be mindful of their mental health and seek help as a preventive measure or if feeling overwhelmed and/or struggling to meet course expectations. Students are encouraged to talk to their instructor for assistance with specific class-related concerns. For additional support and counseling, students are encouraged to contact University Counseling Service (UCS). Information about UCS, including resources and how to schedule an appointment, can be found at counseling.uiowa.edu. Find out more about UI mental health services at: mentalhealth.uiowa.edu.

Basic Needs and Student Support

It can be difficult to maintain focus and be present if you are experiencing challenges with meeting basic needs or navigating personal crisis situations. The Office of the Dean of Students can help. Contact us for one-on-one support, identifying options, and to locate and access basic needs resources (such as food, rent, childcare, etc.).

Student Care and Assistance: 132 IMU, dos-assistance@uiowa.edu, 319-335-1162

Basic Needs info:

- Food Pantry at Iowa

- Clothing Closet
- Basic Needs and Support Resources

Sexual Harassment/Sexual Misconduct and Supportive Measures
The University of Iowa prohibits all forms of sexual harassment, sexual misconduct, and related retaliation. The Policy on Sexual Harassment and Sexual Misconduct governs actions by students, faculty, staff, and visitors. Incidents of sexual harassment or sexual misconduct can be reported to the Office of Civil Rights Compliance or to the Department of Campus Safety. Students impacted by sexual harassment or sexual misconduct may be eligible for academic supportive measures and can learn more by contacting the Office of Civil Rights Compliance. Information about confidential resources can be found on the Office of Civil Rights Compliance website.

Conflict Resolution
The Office of the Ombudsperson is a confidential, impartial, informal, and independent resource for any member of the university community with a problem or concern. The Office of the Ombudsperson offers a safe place to discuss conflicts or concerns. Students are encouraged to reach out for assistance. The office will brainstorm with students to help identify options, answer any questions, and provide referrals to other offices as appropriate. More information about the Office of the Ombudsperson, including how to schedule an appointment, can be found at ombudsperson.org.uiowa.edu.